

Research Article

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Age of onset of first depressive episode requiring hospitalization, clinical characteristics, and response to electroconvulsive therapy

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Abstract

Background. Age at onset of depression is linked to clinical presentation and outcome, but its relevance for electroconvulsive therapy (ECT) remains unclear. We examined whether age at onset of first depressive episode requiring hospitalization predicts ECT response and remission, and whether onset age reflects distinct clinical profiles.

Methods. A nationwide register-based study, including 14,141 patients who received an index ECT series for a depressive episode in Sweden during 2008–2021, with outcome data in the Swedish Quality Register for ECT. Response was “very much/much improved” on the Clinical Global Impression – Improvement scale, and remission was a post-treatment Montgomery-Åsberg Depression Rating Scale-Self report score of 10 or less. Hierarchical logistic regression and causal mediation analyses examined associations of age at onset (early, mid, late), chronological age, baseline severity, psychotic symptoms, and somatic and psychiatric comorbidity with outcome and their mediating role.

Results. Response and remission rates were highest in late-onset depression (84.9%, 60.2%) and lowest in early-onset depression (64.6%, 26.3%). Later onset and older age were independently associated with higher odds of both outcomes, but the onset-age association attenuated substantially after adjustment. The early-onset disadvantage operated mainly through psychiatric comorbidity and treatment complexity, and the late-onset advantage was driven by psychotic features. Associations of age, male sex, and psychotic symptoms with response differed by onset group.

Conclusions. Age of onset of first depressive episode requiring hospitalization was associated with clinical profiles and ECT outcomes, independent of chronological age, suggesting early- and late-onset depression are partly distinct subtypes relevant to treatment planning.

Introduction

Major depressive disorder (MDD) affects an estimated 270 million people worldwide and is more common in women and older adults [1]. Neurobiology, symptomatology, and psychiatric and somatic comorbidities in unipolar depression vary with both current age and age at onset [2, 3].

Early-onset depression (18–29 years) often manifests with pronounced cognitive and emotional symptoms (e.g. irritability, negative self-perception), and high rates of comorbid drug abuse, anxiety, neurodevelopmental, and personality disorders [2–5]. In contrast, a substantial segment of patients suffer a first depressive episode in late life, commonly termed late-onset depression (≥ 65 + years), which is associated with greater executive dysfunction, psychotic symptoms, and somatic comorbidities such as dementia and cardiovascular disease [3, 6–8]. Early-onset depression is further characterized by higher rates of childhood trauma, whereas late-onset depression is more closely linked to a pro-inflammatory state [9]. While older patients often show better electroconvulsive therapy (ECT) outcomes, specific somatic comorbidities prevalent in this group, such as atherosclerosis, diabetes, and COPD, may independently impede ECT outcomes [10]. These age-dependent clinical profiles appear to have a genetic basis. Recent genome-wide association data indicate that early- and late-onset depression have distinct genetic architectures, with early-onset depression enriched for fetal brain signals related to neurodevelopment, and late-onset depression linked to cardiovascular and metabolic risk and cognitive decline [11]. Late-life depression is nonetheless heterogeneous, as it may represent either late-onset or early-onset depression persisting into old age. Because these two groups are not defined in current diagnostic systems, age cut-offs vary across studies, most commonly <30 years for early-onset [12, 13] and 50–65 years for late-onset depression [4, 7].

Although late-life depression may be complex to treat [14, 15], several epidemiological studies suggest that ECT is particularly effective in older patients [16–19]. It remains unclear whether this is due to older age itself, or to correlated factors such as a symptoms profile, comorbidity,



or late-onset depression. Supporting broader age-related differences in depression, psychopharmacological studies report that younger patients are more vulnerable to sustained suicidal ideation after initiation of antidepressants than older adults [20, 21], supporting age-related differences in depression. Beyond age, previous reviews have reaffirmed that psychotic symptoms, melancholic features, shorter episode duration, and higher baseline severity are predictors of a superior ECT response, whereas medication resistance and comorbid personality disorders consistently portend poorer outcomes. [9, 19, 22]. Indeed, a recent causal mediation analysis with 256 participants estimated that 43.1% of the enhanced ECT response seen with older age is explained by clinical mediators, such as psychotic features, psychomotor retardation or agitation, and shorter episode duration, rather than age itself [9, 22, 23], suggesting that the age-related advantage may, in large part, reflect this distinct clinical phenotype rather than a direct effect of aging. Because clinical features currently offer more reliable prognostic value than neuroimaging or genetic markers, understanding these phenotypic relationships is central to precision psychiatry in ECT.

To our knowledge, few previous studies have explicitly examined the influence of age of onset on ECT outcome in depth, although several studies have considered age of onset as one factor in multivariate models. In one study of 110 older patients, the response rate was higher in the late-onset group (86.9%) than in those with earlier onset (<55 years; 67.3%). As there were no significant differences in symptoms or somatic comorbidity between groups, age of onset itself appeared to be a prognostic factor. However, this study included only inpatients aged 55 years and older with severe depression, and did not include chronological age as a predictor in the analyses [24]. Furthermore, while age of onset has been recognized as a phenotypic marker, its role as a mediator is contested. In the mediation study described above, age of onset was not found to mediate the relationship between chronological age and ECT outcome [23]. This discrepancy, together with the genetic overlap between early- and late-onset subtypes, underscores the need for larger-scale investigations to determine if age of onset is a prognostic factor independent of chronological age and other clinical factors.

The primary aim of this study was to investigate whether age of onset of the first depressive episode requiring hospitalization (early, mid, late) and chronological age at treatment predict ECT outcome. We also examined whether any association between age of onset and ECT outcome is independent of other prognostic factors, such as baseline severity or comorbid diagnoses, and whether differential ECT response across early- versus late-onset depression supports etiological heterogeneity between these subgroups. Finally, we assessed whether prognostic clinical factors mediate the association between age of onset and ECT outcomes.

Methods

Participants and study design

This cohort study used three Swedish patient registers. The primary data source was the Swedish National Quality Register for Electroconvulsive Therapy (Q-ECT), which contains detailed information on ECT performed in Sweden [25]. Patients who underwent an index ECT series (≥ 2 sessions/week) between January 1, 2008, and December 31, 2021, for MDD were included (ICD-10 codes F32.1-F32.3 and F33.1-F33.3). Patients were cross-linked using pseudonymized personal identity numbers to the Swedish National Patient Register (NPR) and the National Prescribed Drug Register (NPDR). From these sources, we obtained data on sex, age at the first

diagnosed depressive episode requiring hospitalization, somatic comorbidities, and history of medication use.

Data sources

The Q-ECT began in 2008 as a regional collaboration between psychiatric units in three counties, and became a national register in 2011. Since 2019, coverage has exceeded 90%. Registration is voluntary and requires informed consent. Treatments are recorded as either index series (≥ 2 sessions/week) or maintenance series with less frequent sessions following remission. The register includes data on demographics, illness characteristics, treatment outcomes, and information about the ECT [25].

The NPR and NPDR are mandatory national registers that monitor health care use in Sweden. The NPR, established in 1964, included psychiatric diagnoses in 1973 (86% national coverage), and full national coverage in 1987. Since 2001, it includes specialist outpatient care, but not primary care [26, 27].

The NPDR contains all dispensed prescription drugs in Sweden since 2005 [28].

Outcome measures

The primary outcomes were first a clinician-rated treatment response using the Clinical Global Impression – Improvement (CGI-I) scale, which is a 7-point scale: scores of 1 (“very much improved”) and 2 (“much improved”) were classified as response, and scores 3–7 a non-response [29]. Second, a patient-rated remission from depression using Montgomery-Åsberg Depression Rating Scale – Self report (MADRS-S), a nine-item depression scale in which each item ranges from 0 to 6 (max score, 54), was used to assess remission from depression. Remission was defined as having a total MADRS_S score of 10 or less post-ECT. [30] Ratings were obtained at follow-up interviews after the index ECT had ended, and reflect change from pre-ECT baseline.

Input measures

The main independent variables were chronological age at treatment and age of onset of first depressive episode requiring hospitalization.

Age at treatment was modeled either as a continuous variable or by using a restricted cubic spline with knots at 29 and 65 years, yielding three age segments with 30–64 years as the reference.

Age of onset of first depressive episode requiring hospitalization was derived from the earliest documented inpatient episode for an affective disorder in the NPR, assumed to represent the patient’s first severe depressive episode. Because older ICD versions (ICD-7 to ICD-9) used the umbrella diagnosis “psychosis” for some affective presentations, we also considered these codes. Patients were categorized into early-onset (≤ 29 years), mid-onset (30–64 years) and late-onset (≥ 65 years old) depression, with cut-offs based on previous literature [7, 17–19].

The Clinical Global Impression – Severity (CGI-S) is a 7-point clinician-rated scale ranging from 1 (“normal, not at all ill”) to 7 (“among the most severely ill patients”) [29]. In the Q-ECT, CGI-S is recorded before and after the index series. We used the pre-treatment CGI-S to assess baseline depression severity and included it as a continuous covariate (1–7) in the regression analyses.

To approximate illness complexity, we used prior psychopharmacotherapy as indicating more treatment-resistant depression. Drugs were grouped by Anatomical Therapeutic Chemical (ATC)

codes, each representing a mode of action [31]. We included antidepressants (all N06A group) and other agents that may be prescribed for MDD: antiepileptics (lamotrigine, valproic acid, carbamazepine) and psycholeptics (diazepam, oxazepam, alprazolam, nitrazepam, and flunitrazepam) [32]. One point was assigned per antidepressant ATC code and maximum of one point each for the other drug categories. The resulting total score was treated as a count variable. The data were obtained from the NPDR. Note that this measure did not explicitly capture the number of illness episodes, as a patient may have switched multiple times during the same episode, which would indicate complexity in achieving adequate treatment response.

Data on somatic diseases were obtained from the NPR, where diagnoses are recorded using ICD versions 7–10. To harmonize across ICD revisions, we applied a Swedish adaptation of the Charlson Comorbidity Index (CCI) for register-based research [33], which groups equivalent codes for the same condition across ICD 7–10 into single categories [33]. Somatic comorbidity was treated as a categorical variable with three levels: no comorbidity, one comorbidity, and two or more comorbidities. The diagnoses included in the comorbidity variable are presented in Table 1.

Data on comorbid psychiatric disorders were obtained from the NPR. Anxiety disorders, personality disorders, substance use disorders, and neurodevelopmental disorders were included as separate dichotomous variables, as these commonly co-occur with MDD, particularly in younger people [34, 35].

To adjust for time lived that was not covered by the register data, we included the variable Proportion of life not in observable registers as covariate in all main analyses. The variable was stratified into three categories: 1) Fully observed, 2) 0–29 years unobserved, and 3) 30–64 years unobserved. These year-intervals were derived from the age of onset categories, indicating whether the patient could have either one or two onset categories potentially misclassified due to years lived outside of observable register time.

Statistical analysis

Data were managed and analyzed using SAS v. 9.4 and R v. 4.4.3. Frequencies and means were used for descriptive statistics of key variables. Hierarchical logistic regression, with ECT response and remission from depression as the outcomes, was performed to assess the independent and joint role of age of onset and other potentially influential factors. In this stepwise approach, variables are sequentially entered in blocks, allowing evaluation of incremental model improvement and the added predictive value of age of onset when successively adjusting for age at treatment and other covariate groups [36]. Second, we stratified the sample by age-of-onset group and ran separate logistic regression models to identify predictors of ECT response within each category, and assessed interaction by onset group for each variable. Finally, causal mediation analyses examined whether somatic and psychiatric comorbidity, psychotic features, illness severity, and number of previous medications mediated the association between age of onset and ECT outcomes. Two exposure contrasts were tested, late onset (≥ 65) versus mid/early onset, and early onset (< 30) versus mid/late onset, for both response and remission. Multi-mediator analyses were conducted using the R package *CMAverse* (regression-based imputation), modeling all five mediators simultaneously and decomposing the total effect into direct and indirect components. Single-mediator analyses using the *mediation* package were conducted in parallel to estimate the contribution of each mediator individually. Both approaches adjusted for age, age², sex, and

Table 1. Patient and treatment characteristics of the 14,141 patients included in the study

Variable	Early onset ≤ 29 years	Mid onset 30–64 years	Late onset ≥ 65 years
Number of patients, <i>n</i>	3,912	7,585	2,644
Response rate, <i>n</i> (%)	2,528 (64.6)	5,990 (79.0)	2,244 (84.9)
Remission rate, <i>n</i> (%)	554 (26.3)	1,867 (42.7)	885 (60.2)
Demographic information			
Sex, male, <i>n</i> (%)	1,524 (39.0)	3,343 (44.1)	1036 (39.2)
Mean age (range)	34.5 (13–78)	55.8 (30–96)	76.0 (65–99)
Indication for ECT			
Psychotic depression, <i>n</i> (%)	652 (16.7)	1,852 (24.4)	952 (36.0)
Mean CGI-S prior to ECT, M(SD)	5.2 (0.8)	5.1 (0.8)	5.2 (0.8)
Mean MADRS-S prior to ECT, M(SD)	35.8 (8.4)	33.6 (8.9)	30.3 (9.7)
Previous treatments			
Previous ECT, <i>n</i> (%)	919 (28.3)	2,021 (33.2)	520 (24.8)
Mean number of medications	5.5 (3.1)	5.2 (2.8)	4.5 (2.5)
The received ECT series			
Mean number of sessions in the series, M(SD)	7.8 (3.1)	7.7 (3.1)	7.4 (3.0)
Mean charge (mC), M(SD)	281.1 (130.9)	361.4 (143.7)	429.6 (153.9)
Mean seizure duration, M(SD)	52.7 (27.6)	48.6 (22.5)	48.3 (21.9)
Electrode placement	3,503 (89.9)	6,766 (89.5)	2,348 (89.2)
- Right unilateral stimulation, <i>n</i> (%)			
- Bitemporal stimulation, <i>n</i> (%)	346 (8.9)	677 (9.0)	246 (9.4)
- Other, <i>n</i> (%)	48 (1.2)	117 (1.5)	38 (1.5)
Pulse width	0.50 (0.1)	0.52 (0.1)	0.54 (0.1)
Psychiatric comorbidities			
Anxiety, <i>n</i> (%)	1,833 (46.9)	3,408 (44.9)	1,156 (43.7)
Neurodevelopmental disorder, <i>n</i> (%)	340 (8.7)	127 (1.7)	2 (0.1)
Personality disorder, <i>n</i> (%)	355 (9.1)	246 (3.2)	20 (0.8)
Substance disorder, <i>n</i> (%)	594 (15.2)	866 (11.4)	133 (5.0)
Somatic comorbidities			
Diabetes	167 (4.3)	648 (8.5)	254 (9.6)
Dementia	9 (0.2)	95 (1.3)	124 (4.7)
Cerebrovascular disease	54 (1.4)	344 (4.5)	289 (10.9)
Hemiplegia	42 (1.1)	67 (0.9)	31 (1.2)
Severe kidney disease	74 (1.9)	131 (1.7)	77 (2.9)
Severe liver disease	13 (0.3)	27 (0.4)	8 (0.3)
Myocardial infarction	26 (0.7)	227 (3.0)	151 (5.7)
Peptic ulcer disease	65 (1.7)	215 (2.8)	109 (4.1)
Congestive heart failure	32 (0.8)	177 (2.3)	135 (5.1)
Chronic obstructive pulmonary disease	48 (1.2)	193 (2.5)	78 (3.0)
Metastatic cancer	10 (0.3)	85 (1.1)	47 (1.8)

Continued

Table 1. *Continued*

Variable	Early onset ≤29 years	Mid onset 30–64 years	Late onset ≥65 years
Peripheral vascular disease	24 (0.6)	118 (1.6)	98 (3.7)
Rheumatic disease	134 (3.4)	573 (7.6)	318 (12.0)

Notes: Values are either the number of positive cases (*n*) and their percentage share of the total sample included in the study (%) or the mean value, depending on what is relevant for the specific variable.

Response rate was assessed using the seven-item scale Clinical Global Impression – Improvement (CGI-I) scale. Scores of 1 (“Very much improved”) and 2 (“Much improved”) were assessed as “response.” Remission was assessed using the 54-point Self-Assessed Montgomery-Åsberg Depression Rating Scale (MADRS-S). MADRS-S is filled out by the patients before and after the completed ECT series. A score of <10 points was assessed as “remission.” Clinical Global Impression – Severity (CGI-S) scale is a seven-item scale describing the objective severity of the disease assessed by the clinician before and after the ECT. Mean MADRS-S and CGI-S before ECT were presented in order to describe the severity of depression. Number of medications was obtained from the National Prescribed Drug Register using the Anatomical Therapeutic Chemical (ATC) code. The psychiatric and somatic comorbidities were obtained from the National Patient Register, where they are registered using International Statistical Classification of Diseases and Health Related Problems, revisions 7–10 (ICD-7 to ICD-10). The revisions were unified and diagnostic codes classified using the Charlson Comorbidity Index (CCI). Neurodevelopmental disorders include autism spectrum disorder and/or ADHD. Data on the number of sessions in the series, applied stimulation energy, and seizure duration were obtained from the Swedish Quality Register for the ECT (Q-ECT). The Q-ECT collects data on stimulation energy applied during the first session of the series and seizure duration measured during the last session of the series.

register observation completeness, with 1000 bootstrap resamples throughout.

In addition, a sensitivity analysis was performed, where the time since first antidepressant dispensation was obtained in those patients who had their first depressive episode requiring hospitalization during the years 2019–2021. The first antidepressant dispensation was considered a proxy for the patient’s first depressive episode. This time-lag was subsequently used in a Monte Carlo simulation applied to the entire dataset, on which the main analysis was re-run, to approximate the difference in associations between first depressive episode requiring hospitalization and first ever depressive episode. This analysis is detailed in the [Supplementary Materials](#).

Ethical considerations

Participation in the Q-ECT is voluntary, and consent may be withdrawn at any time. The study forms part of a larger project approved by the Regional Ethical Review Board in Uppsala (registration number 2014/174) and the Swedish Ethical Review Authority (2021-03815).

Results

Patient characteristics and response and remission rates to ECT

Between January 1, 2008 and December 31, 2021, 14,141 patients in Sweden received an index series for MDD registered in the Q-ECT and had a non-missing CGI-I rating.

Most patients (7,585; 53.6%) had mid-onset depression (30–64 years), 3,912 (27.7%) had early-onset depression (≤29 years), and 2,644 (18.7%) had late-onset depression (≥65 years). Mean age at treatment was 34.5, 55.8, and 76.0 years in the early-, mid-, and late-onset groups, respectively. Response rate to ECT (CGI-I score 1–2) was highest in the late-onset group (84.9%) and the lowest in the early-onset group (64.6%). Psychosis was present in 16.7% of early-onset patients and 36.0% of late-onset patients. Early-onset patients had tried the most medications before ECT (mean 5.5 versus 4.5 in

late-onset). All included psychiatric comorbidities were most prevalent among the early-onset and least prevalent in the late-onset group. Further details are provided in [Table 1](#).

Hierarchical binomial logistic regression of predictors of response and remission

As shown in [Table 2](#), when only age of onset and register observation completeness were considered (Block I), early-onset depression was associated with a lower likelihood of both ECT response and remission compared with mid-onset (both OR 0.65, $p < .001$), whereas late-onset depression was associated with a higher likelihood of response (OR 1.28, $p < .001$) and remission (OR 1.67, $p < .001$). Longer periods of life unobserved in the registers were strongly associated with better outcomes in Block I, but these associations became non-significant once chronological age was entered in Block II, suggesting they largely reflected age effects.

When chronological age and sex were added (Block II), chronological age was a strong independent predictor of both outcomes. The early-onset disadvantage remained significant, while the late-onset advantage was no longer significant for response, although only marginally, but remained significant for remission (OR 1.37, $p < .001$).

After adding depression characteristics (Block III), the early-onset association attenuated further for response (OR 0.86, $p = .016$) and became non-significant for remission. Late-onset depression was non-significant for both outcomes at this stage.

Adding comorbid illnesses in Block IV produced a further model improvement. Early- and late-onset associations were non-significant for both outcomes, though the late-onset remission association approached significance (OR 1.17, $p = .055$). Two or more somatic comorbidities, anxiety disorder, substance use disorder, and personality disorder were each independently associated with poorer outcomes. Chronological age remained a significant predictor throughout all blocks.

Prognostic factors of response and remission, divided by age of onset group

As can be seen in [Table 3](#), prognostic factors for ECT response and remission differed across age-of-onset groups. In the early-onset group, psychiatric comorbidities were prominent negative predictors of both outcomes, including anxiety syndrome, substance use disorder, and personality disorder, whereas these associations were largely absent in mid- and late-onset groups. Psychotic symptoms, baseline illness severity, and number of prior medications predicted response across all onset groups; for remission, illness severity was significant only in the mid-onset group. Somatic comorbidities (two or more) predicted poorer response in both mid- and late-onset groups, but for remission only in the mid-onset group.

There was a significant interaction between age of onset and chronological age for response, such that the positive association between older age and ECT response weakened with later onset. For sex, men in the early-onset group were more likely than women to respond and achieve remission, whereas the opposite pattern was observed in the mid-onset group; this interaction was significant for both response and remission. Psychotic symptoms were a significantly stronger predictor of both response and remission in early-onset than late-onset depression. For remission, a significant interaction was also observed for personality disorder, driven by a strong negative association in early onset that attenuated across groups.

Table 2. Hierarchical binomial logistic regression of predictors of response to ECT and remission from depression, including odds ratios for baseline predictors

Variable	Response		Remission	
	OR (95% CI)	P	OR (95% CI)	P
BLOCK I, age of onset				
Response				
Block χ^2 (4) = 536.01, $P < .001$. Nagelkerke R^2 = 0.056				
Remission				
Block χ^2 (4) = 519.01, $P < .001$. Nagelkerke R^2 = 0.085				
<30 years unobserved	1.72 (1.56–1.91)	<.001	1.78 (1.58–2.02)	<.001
≥30 years unobserved	1.82 (1.55–2.14)	<.001	2.11 (1.76–2.53)	<.001
Early onset depression	0.65 (0.59–0.72)	<.001	0.65 (0.57–0.75)	<.001
Late onset depression	1.28 (1.12–1.47)	<.001	1.67 (1.46–1.97)	<.001
BLOCK II, chronological age and sex				
Response				
Block χ^2 (7) = 592.99, $P < .001$. Nagelkerke R^2 = 0.062, Model change χ^2 (3) = 56.98, $P < .001$				
Remission				
Block χ^2 (7) = 575.97, $P < .001$. Nagelkerke R^2 = 0.094, Model change χ^2 (3) = 56.96, $P < .001$				
<30 years unobserved	1.02 (0.86–1.21)	.851	1.00 (0.81–1.24)	0.993
≥30 years unobserved	0.88 (0.67–1.17)	.391	0.85 (0.63–1.16)	0.303
Early onset depression	0.79 (0.70–0.88)	<.001	0.78 (0.68–0.90)	<.001
Late onset depression	1.12 (0.97–1.31)	.133	1.37 (1.18–1.60)	<.001
Age, spline ≤29 years	6.16 (3.81–9.98)	<.001	7.48 (4.01–14.00)	<.001
Age, spline 65 + years	1.84 (1.37–2.48)	<.001	2.46 (1.81–3.35)	<.001
Sex, female	1.05 (0.96–1.13)	0.278	1.08 (0.98–1.19)	.113
BLOCK III, depression characteristics				
Response				
Block χ^2 (10) = 1184.37, $P < .001$. Nagelkerke R^2 = 0.123, Model change χ^2 (3) = 938.67, $P < .001$				
Remission				
Block χ^2 (10) = 1010.08, $P < .001$. Nagelkerke R^2 = 0.165, Model change χ^2 (3) = 705.39, $P < .001$				
<30 years unobserved	0.92 (0.77–1.10)	.378	0.88 (0.70–1.10)	.256
≥30 years unobserved	0.91 (0.68–1.21)	.506	0.80 (0.58–1.10)	.166
Early onset depression	0.86 (0.76–0.97)	.016	0.91 (0.78–1.06)	.207
Late onset depression	1.01 (0.86–1.19)	.882	1.20 (1.03–1.41)	.023
Age, spline ≤29 years	8.97 (5.43–14.9)	<.001	12.4 (6.48–24.0)	<.001
Age, spline 65 + years	1.57 (1.16–2.13)	<.001	2.46 (1.79–3.40)	<.001
Sex, female	0.95 (0.87–1.03)	.239	0.96 (0.87–1.06)	.474
Psychotic symptoms	1.74 (1.55–1.96)	<.001	1.80 (1.60–2.02)	<.001
Illness severity at baseline (CGI-S)	1.45 (1.38–1.53)	<.001	1.13 (1.06–1.21)	<.001
Number of previous medication types	0.92 (0.91–0.93)	<.001	0.88 (0.86–0.89)	<.001
BLOCK IV, comorbid illnesses				
Response				
Block χ^2 (16) = 1214.07, $P < .001$. Nagelkerke R^2 = 0.126, Model change χ^2 (5) = 29.7, $P < .001$				
Remission				
Block χ^2 (16) = 1057.61, $P < .001$. Nagelkerke R^2 = 0.169, Model change χ^2 (5) = 47.53, $P < .001$				
<30 years unobserved	0.91 (0.86–1.09)	.341	0.86 (0.69–1.08)	.215
≥30 years unobserved	0.89 (0.66–1.19)	.417	0.78 (0.56–1.08)	.121
Early onset depression	0.90 (0.79–1.01)	.079	0.98 (0.84–1.15)	.469
Late onset depression	0.99 (0.84–1.16)	.926	1.17 (1.00–1.37)	.055
Age, spline ≤29 years	10.2 (6.13–17.1)	<.001	15.7 (8.09–30.6)	<.001
Age, spline 65 + years	1.70 (1.25–2.33)	<.001	2.65 (1.91–3.68)	<.001

Continued

Table 2. Continued

Variable	Response		Remission	
	OR (95% CI)	P	OR (95% CI)	P
Sex, female	0.95 (0.88–1.04)	0.266	0.97 (0.8–1.07)	.536
Psychotic symptoms	1.73 (1.54–1.95)	<.001	1.79 (1.59–2.01)	<.001
Illness severity at baseline (CGI-S)	1.45 (1.38–1.53)	<.001	1.13 (1.07–1.21)	<.001
Number of previous medication types	0.93 (0.91–0.94)	<.001	0.89 (0.87–0.91)	<.001
Somatic comorbidities				
- One	0.99 (0.88–1.11)	.788	0.94 (0.83–1.06)	.252
- Two or more	0.70 (0.59–0.83)	<.001	0.68 (0.55–0.83)	<.001
Anxiety disorder	0.88 (0.80–0.97)	.010	0.88 (0.79–0.98)	.019
Substance use disorder	0.83 (0.73–0.95)	.008	0.73 (0.61–0.86)	<.001
Neurodevelopmental disorder	0.86 (0.69–1.06)	.157	0.88 (0.64–1.19)	.395
Personality syndrome	0.81 (0.67–0.98)	.031	0.52 (0.39–0.69)	<.001

Notes: Analyzed sample: 14,141 patients.

Values are odds ratios (OR) with 95% confidence intervals (CI) in the parentheses. Cut-off age for early onset depression is 0 to 29 years, and late onset depression is 65 years or older. Both onset groups are compared with mid-age onset depression (30–64 years) as reference category. ICD codes F32.3 or F33.3 were assessed as psychotic depression. Severity was assessed using the Clinical Global Impression – Severity scale (1–7). Number of medications was obtained from the National Prescribed Drug Register using the Anatomical Therapeutic Chemical (ATC) code. The presence of somatic comorbidities was obtained from the National Patient Register and classified using the Charlson Comorbidity Index (CCI). All models were also adjusted for electrode placement, dose charge, and pulse width (not shown). Adjusted models were not statistically significantly different from unadjusted models. Variables in bold were added in the current block.

Causal mediation analysis of clinical mediators in the pathway from age of onset of first depressive episode requiring hospitalization, to response and remission

Causal mediation analyses examined whether the association between age of onset and ECT outcomes operated through somatic and psychiatric comorbidity, psychotic features, illness severity at initiation of ECT, and number of previous medications.

As shown in Table 4, for late onset (≥ 65) versus mid- and early-onset, the total effect favored late onset for both response (OR 1.21, 95% CI 1.02–1.42, $p = .024$) and remission (OR 1.31, 95% CI 1.13–1.54, $p < .001$). In the multi-mediator analysis, the joint indirect effect was significant for both outcomes (OR 1.14 and 1.17, both $p < .001$), accounting for approximately 67% of the total effect on response and 45% on remission. The direct effect was non-significant for response ($p = .43$) but significant for remission (OR 1.17, $p = .044$). Single-mediator analyses confirmed significant indirect effects for all five mediators (all $p \leq .006$), with number of previous medications showing the largest proportion mediated (response 53%, remission 30%). The direct effect was non-significant for response across all mediators, but remained significant for remission (all $p \leq .004$), indicating a residual direct pathway not explained by any individual mediator regarding remission.

For early onset (< 30) versus mid- and late-onset, the total effect disfavored early onset (response OR 0.83, 95% CI 0.74–0.94, $p < .001$; remission OR 0.79, 95% CI 0.67–0.92, $p = .004$), and the direct effect was non-significant in both the multi-mediator ($p = 0.33$ and $p = 0.80$) and single-mediator analyses, suggesting the disadvantage associated with early onset operated primarily through the clinical mediators. Significant indirect effects were observed for somatic and psychiatric comorbidity, psychotic features, and number of medications (all $p < .001$). Illness severity was an exception, with a non-significant indirect effect for both outcomes ($p = .082$ and $p = .178$) and a point estimate in the opposite direction to the total effect. Number of previous medications again

showed the highest proportion mediated (response 45%, remission 57%) (Table 4).

Discussion

This study investigated the independent and joint associations of age of onset of first depressive episode requiring hospitalization and chronological age at treatment with ECT response and remission from depression in a nationwide cohort of 14,141 patients, and whether clinical and comorbidity factors account for these associations. Both age of onset and chronological age were independently associated with outcome: response and remission rates increased with later onset, from 64.6% and 26.3% in early-onset, to 79% and 42.7% in mid-onset, to 84.9% and 60.2% in late-onset. Increasing chronological age, up to approximately 65 years, was likewise associated with a greater likelihood of a positive outcome, and this was a distinct effect rather than a re-statement of the effect of age of onset. Chronological age remained a robust predictor throughout the fully adjusted models, whereas the association between age of onset and outcome attenuated substantially, or almost entirely, once depression characteristics and comorbid illness were accounted for. The causal mediation analyses indicate that distinct clinical pathways underlie the early-onset disadvantage and the late-onset advantage.

Within early- and mid-onset groups, higher chronological age at treatment was associated with better outcome, an association absent in late-onset depression. Previous research on this topic has been limited and often inconsistent. One of the few previous studies explicitly examining age of onset of depression similarly reported that late-onset depression was associated with better ECT response [24], though that study did not include age at treatment as co-predictor in the same model.

Our observation that early-onset depressive episode requiring hospitalization was associated with poorer ECT outcomes aligns

Table 3. Prognostic factors of response to ECT and remission from depression, divided by age of onset group, as well as interaction terms of age of onset by each factor

Variable	OR (95% CI)			Interaction	
	Early onset ≤ 29 years	Mid onset 30–64 years	Late onset ≥ 65 years	Wald χ^2 (df)	P
Response					
Life observed (ref: Fully observed)					
<30 years unobserved	0.87 (0.63–1.21)	1.08 (0.88–1.33)	–	1.75 (1)	.186
≥30 years unobserved	–	0.97 (0.66–1.43)	0.98 (0.70–1.37)	1.18 (1)	.278
Age	1.03 (1.02–1.05)	1.02 (1.01–1.03)	0.99 (0.97–1.02)	8.02 (1)	.005
Sex, male, <i>n</i> (%)	1.17 (1.01–1.36)	0.86 (0.77–0.97)	0.85 (0.68–1.07)	8.04 (1)	.005
Somatic comorbidities					
- One	1.10 (0.86–1.39)	0.97 (0.84–1.12)	0.94 (0.73–1.21)	0.73 (1)	.392
- Two or more	0.94 (0.58–1.55)	0.69 (0.55–0.87)	0.61 (0.45–0.83)	2.05 (1)	.152
Anxiety syndrome	0.79 (0.65–0.94)	0.93 (0.82–1.06)	0.89 (0.71–1.13)	1.09 (1)	.297
Neurodevelopmental disorder, <i>n</i> (%)	0.75 (0.57–0.99)	0.90 (0.58–1.33)	0.55 (0.01–22.88)	0.23 (1)	.631
Personality disorder, <i>n</i> (%)	0.77 (0.59–1.02)	0.87 (0.64–1.19)	0.43 (0.16–1.27)	0.17 (1)	.677
Substance disorder, <i>n</i> (%)	0.73 (0.58–0.92)	0.84 (0.70–1.02)	1.39 (0.81–2.51)	2.03 (1)	.154
Psychotic symptoms	2.30 (1.85–2.90)	1.59 (1.35–1.88)	1.42 (1.09–1.85)	9.21 (1)	.002
Illness severity at baseline (CGI-S)	1.36 (1.25–1.48)	1.54 (1.43–1.66)	1.40 (1.22–1.62)	1.29 (1)	.255
Number of previous medication types	0.94 (0.92–0.97)	0.92 (0.90–0.94)	0.93 (0.89–0.98)	1.09 (1)	.296
Remission					
Life observed (ref: Fully observed)					
<30 years unobserved	0.71 (0.45–1.15)	0.95 (0.75–1.20)	–	4.35 (1)	.037
≥30 years unobserved	–	0.88 (0.57–1.35)	1.02 (0.73–1.44)	0.82 (1)	.365
Age	1.04 (1.03–1.06)	1.03 (1.02–1.04)	0.98 (0.95–1.00)	3.61 (1)	.057
Sex, male, <i>n</i> (%)	1.24 (1.00–1.54)	0.93 (0.81–1.05)	0.84 (0.67–1.05)	5.19 (1)	.023
Somatic comorbidities					
- One	0.75 (0.52–1.05)	0.90 (0.76–1.05)	1.19 (0.93–1.52)	4.92 (1)	.027
- Two or more	0.60 (0.30–1.16)	0.60 (0.45–0.80)	0.84 (0.60–1.18)	1.47 (1)	.226
Anxiety syndrome	0.62 (0.48–0.81)	0.96 (0.83–1.10)	0.91 (0.72–1.15)	3.23 (1)	.072
Neurodevelopmental disorder, <i>n</i> (%)	0.84 (0.55–1.25)	0.57 (0.31–1.00)	–	3.46 (1)	.063
Personality disorder, <i>n</i> (%)	0.33 (0.20–0.54)	0.63 (0.42–0.94)	1.52 (0.48–5.73)	4.74 (1)	.029
Substance disorder, <i>n</i> (%)	0.57 (0.41–0.79)	0.75 (0.60–0.94)	0.85 (0.51–1.41)	0.65 (1)	.419
Psychotic symptoms	2.86 (2.20–3.72)	1.68 (1.44–1.96)	1.37 (1.06–1.76)	15.22 (1)	<.001
Illness severity at baseline (CGI-S)	1.04 (0.92–1.19)	1.22 (1.12–1.32)	1.05 (0.91–1.22)	0.06 (1)	.805
Number of previous medication types	0.91 (0.88–0.95)	0.89 (0.86–0.91)	0.87 (0.83–0.91)	2.78 (1)	.095

Note: Numbers marked in bold are significant at $P < .05$ level according to the penalized maximum likelihood estimates. All models were also adjusted for electrode placement, dose charge, and pulse width (not shown). Adjusted models were not statistically significantly different from unadjusted models. Life observed, portion of life observed in the registers.

with prior evidence that early-onset depression is more often accompanied by psychiatric comorbidity [13, 37], which is itself associated with reduced treatment response [38, 39]. In our cohort, early-onset patients showed markedly higher rates of anxiety, personality syndromes, and substance use disorder, alongside a greater number of previously dispensed medication types, which is consistent with a more comorbid, treatment-resistant phenotype. Late-onset depression, conversely, has been more closely linked to melancholic and psychotic presentations [23, 40, 41]. This

age-dependent phenotypic heterogeneity matches the concept of distinct, partly age-dependent forms of depression proposed in recent narrative reviews of ECT predictors [9, 22]. It is further supported by recent genome-wide association data, where early- and late-onset depression shared only a moderate degree of genetic overlap. Early-onset depression is more heritable than late-onset depression, and the risk for early-onset depression is concentrated in genes active during fetal brain development rather than in the adult brain, pointing to a neurodevelopmental origin. Early-onset

Table 4. Causal mediator analyses of clinical mediators in the path of onset of first depressive episode requiring hospitalization to response to ECT or remission from depression

Mediator	NIE (95% CI)	<i>p</i> (NIE)	NDE (95% CI)	<i>p</i> (NDE)	Total	% mediated
Predictor: late-onset versus early- and mid-onset. Outcome: Response to ECT						
Overall mediation (odds ratio scale)	1.142 [1.107, 1.185]	<.001	1.069 [0.91, 1.25]	.430	1.207	66.8
Somatic comorbidity	0.004 [0.002, 0.006]	<.001	0.018 [−0.008, 0.042]	.170	0.022	18.2
Psychiatric comorbidity	0.005 [0.003, 0.008]	<.001	0.017 [−0.009, 0.041]	.196	0.022	24.3
Psychotic features	0.01 [0.006, 0.013]	<.001	0.012 [−0.015, 0.036]	.376	0.021	45.3
Illness severity (CGI-S)	0.007 [0.003, 0.011]	<.001	0.016 [−0.01, 0.04]	.226	0.023	30.2
Number of previous medications	0.012 [0.009, 0.015]	<.001	0.01 [−0.016, 0.035]	.430	0.022	53.2
Predictor: late-onset versus early- and mid-onset depression. Outcome: Remission from depression						
Overall mediation (odds ratio scale)	1.168 [1.125, 1.224]	<.001	1.169 [1.005, 1.357]	.044	1.308	45.3
Somatic comorbidity	0.006 [0.002, 0.01]	<.001	0.07 [0.035, 0.107]	.002	0.076	7.4
Psychiatric comorbidity	0.01 [0.006, 0.015]	<.001	0.065 [0.029, 0.101]	.002	0.076	13.8
Psychotic features	0.017 [0.008, 0.023]	<.001	0.059 [0.023, 0.095]	.002	0.076	22.0
Illness severity (CGI-S)	0.005 [0.001, 0.009]	.006	0.067 [0.03, 0.103]	<.001	0.072	6.6
Number of previous medications	0.022 [0.015, 0.03]	<.001	0.053 [0.017, 0.088]	.004	0.075	29.8
Predictor: early-onset versus mid- and late-onset depression. Outcome: Response to ECT						
Overall mediation (odds ratio scale)	0.903 [0.873, 0.937]	<.001	0.903 [0.873, 0.937]	.330	0.831	64.4
Somatic comorbidity	−0.003 [−0.005, −0.002]	<.001	−0.038 [−0.058, −0.016]	<.001	−0.041	8.0
Psychiatric comorbidity	−0.014 [−0.017, −0.01]	<.001	−0.028 [−0.048, −0.006]	.014	−0.041	32.9
Psychotic features	−0.005 [−0.009, −0.002]	<.001	−0.036 [−0.056, −0.015]	<.001	−0.041	11.6
Illness severity (CGI-S)	0.003 [0, 0.006]	.082	−0.041 [−0.063, −0.02]	<.001	−0.038	−8.1
Number of previous medications	−0.018 [−0.022, −0.014]	<.001	−0.022 [−0.042, −0.001]	.048	−0.040	45.0
Predictor: early-onset versus mid- and late-onset depression. Outcome: Remission from depression						
Overall mediation (odds ratio scale)	0.847 [0.809, 0.881]	<.001	1.021 [0.875, 1.18]	.798	0.787	109.8
Somatic comorbidity	−0.006 [−0.01, −0.004]	<.001	−0.051 [−0.083, −0.018]	.002	−0.058	10.9
Psychiatric comorbidity	−0.022 [−0.028, −0.017]	<.001	−0.035 [−0.068, −0.003]	.026	−0.057	38.9
Psychotic features	−0.009 [−0.017, −0.005]	<.001	−0.048 [−0.08, −0.016]	.002	−0.057	16.1
Illness severity (CGI-S)	0.002 [−0.001, 0.006]	.178	−0.058 [−0.092, −0.023]	<.001	−0.056	−4.1
Number of previous medications	−0.033 [−0.042, −0.026]	<.001	−0.026 [−0.057, 0.006]	.144	−0.059	56.7

Note: Both multiple mediator analyses and single mediator analyses are included.

Multi-mediator causal mediation analysis (regression-based imputation, presented on Odds Ratio scale), with mediators modeled in parallel, and single-mediator causal mediation analyses (risk difference scale). Bootstrapped 95% CI (1000 simulations). NIE, natural indirect effect; NDE, natural direct effect. Covariates: age, age², sex, and observation completeness.

depression also shares more genetic risk factors with other psychiatric disorders, while late-onset depression is linked to cardiovascular disease [11]. These genetic findings reinforce the clinical picture, where early- and late-onset depression appear to be partly distinct conditions, with different biological underpinnings, rather than simply milder or more severe presentations of the same underlying illness. Our results extend previous study that similarly reported a higher ECT response rate in late-onset than earlier-onset depression but, restricted to inpatients aged 55 years and older, did not model chronological age as co-predictor [24]. Our sample, about ten times larger than the combined samples of previous studies of age of onset in ECT, allowed both variables to be modeled jointly. Causal mediation analyses indicated that the early-onset disadvantage reflected both a greater burden of psychiatric comorbidity and history of treatment complexity (number of previously dispensed medication types) rather than illness severity, with the

direct (unmediated) effect of early onset non-significant for both outcomes once all five mediators were modeled jointly. Conversely, the late-onset advantage was driven chiefly by a higher prevalence of psychotic features, with fewer previous medications and lower psychiatric comorbidity contributing further. With the five mediators included, the direct effect on response was non-significant, while they left a small but statistically significant direct effect of late onset on remission unexplained. This degree of mediation is broadly comparable to that of the chronological age advantage previously found to be explained by psychotic features, psychomotor retardation and agitation, and episode duration, but not age of onset [23]. Our lack of data on psychomotor retardation and episode duration may in part explain why chronological age remained an independent predictor of outcome in our fully adjusted models even after accounting for the symptoms and comorbidity data available to us.

Psychotic symptoms illustrate the heterogeneity between onset groups particularly clearly. Although psychotic depression was more prevalent in late-onset depression and has previously been considered a hallmark of late-onset and late-life depression [23, 40–44], its predictive association with ECT response and remission was, paradoxically, considerably stronger in early- than in late-onset of first depressive episode requiring hospitalization. This contrasts prior literature suggesting that the predictive value of psychotic symptoms is, if anything, stronger in older patients [9, 22]. Other onset-dependent differences point in the same direction, with male sex associated with better outcomes in early-onset and sex differences attenuating in mid- and late-onset. Comorbid personality syndromes were associated with markedly poorer remission specifically in early-onset age of depressive episode requiring hospitalization, an association that attenuated with later onset. Somatic comorbidity showed an inconsistent pattern. The overall burden of somatic comorbidity was greater in late-onset first depressive episode requiring hospitalization and contributed a favorable indirect effect to the late-onset advantage in the mediation analysis. Yet, in the onset-stratified models, having two or more somatic comorbidities independently predicted poorer, not better, ECT outcome within the mid- and late onset strata specifically. Although this discrepancy was not resolved by the present analyses, the within-stratum finding is consistent with other recent evidence identifying specific somatic conditions as independent risk factors for ECT non-remission [10]. With specific regard to the “vascular depression” hypothesis, which proposes that the psychotic, melancholic, and psychomotor features common in late-onset depression reflect underlying cerebrovascular pathology [45], we found no association between cerebrovascular disease specifically and ECT outcome in sensitivity analyses, which is consistent with a previous study reporting no association between ECT response and cognitive decline or neurological degeneration [24]. It may therefore be that cerebrovascular disease contribute to the development of the late-onset phenotype without directly influencing ECT responsiveness. The residual, unmediated component of the late-onset remission advantage may instead reflect age-related neuroplastic or neuroinflammatory mechanisms. For instance, ECT-induced reductions in microglial-derived inflammatory markers have been linked to increases in hippocampal volume and to symptom improvement [9, 22, 46, 47], though this remains speculative in the absence of relevant biomarker data in our register.

Strengths of this study include its large study sample and broad inclusion criteria across all adult age groups, which enhance the generalizability of these findings. Although including correlated variables such as age, age of onset, and number of previous medications in the same models raises the possibility of multicollinearity, multicollinearity diagnostics did not indicate problematic levels. Another limitation is the absence of data on duration of the current depressive episode and psychomotor retardation, which are not recorded in the Q-ECT. As with all register-based studies, selection and information bias cannot be excluded, since analyses are restricted to information available in the registers. A more fundamental limitation concerns identifying the true age of onset of depression from register data. We operationalized this as age at first depressive episode requiring hospitalization, supplemented by a sensitivity analysis using the date of first recorded antidepressant dispensation as a proxy for true onset. While the reliability of Swedish register data is generally high, this remains an approximation of the actual first episode of depression, and we cannot rule out that our findings would differ under an alternative operationalization.

This study shows that early- and late-onset depression differ not only in ECT outcome but also in clinical profile and, plausibly, in underlying pathophysiology, with most of the early-onset disadvantage, and part of the late-onset advantage, explained by differences in psychiatric comorbidity, treatment complexity, and psychotic symptomatology. Chronological age showed an independent, non-linear association with outcome, improving the likelihood of response and remission up to approximately 65 years. If aging itself enhances susceptibility to the antidepressant effects of ECT, depressive episodes may become increasingly responsive to ECT as patients grow older. Clinically, this implies that a poor ECT response earlier in life should not preclude renewed consideration of ECT for a later depressive episode in the same patient.

Supplementary material. The supplementary material for this article can be found at <http://doi.org/10.1192/j.eurpsy.2026.12248>.

Data availability statement. The data that support the findings of this study are available from Axel Nordenskjöld upon reasonable request.

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Competing interests. The authors have no conflict of interest to declare.

Ethical considerations. The study was approved by the Regional ethics vetting board of Uppsala (registration number 2014/174) and the Swedish Ethical Review Authority (2021-03815). Patients could choose not to have data on ECT included in the Swedish National Quality Registry.

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